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Bid-ask bounds for option prices: the two-tail distortion model

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We model the bid-ask spreads of call and put options by a two-tail distortion (2TD) of a reference probability distribution. The model applies the Choquet pricing approach with no-arbitrage restrictions, requiring a duality relationship between the capacities pricing long and short positions of call and put options. Moreover, the put-call parity relationship requires that the sum of bid-ask spreads of call and put options with the same strike be invariant across the strikes. We calibrate the 2TD model with a simple Sugeno distortion on a sample of two months daily data for three stock indexes and three different reference models and show that the 2TD generally provides a better fit to the data than the standard distortion of one tail only. Moreover, the estimate of the distortion parameter happens to be very similar across the different models.

Keywords: Fuzzy measures; Choquet pricing; Bid-ask spreads; Option pricing

1. Introduction

In this paper, we investigate the relationship between the bid-ask prices of options and those of the corresponding underlying assets in an arbitrage-free setting. We address the problem in the setting of probability distortion models. The family of models that rely on probability distortions includes Jouini and Kallal (1995), Chateauneuf (1996), Cherubini (1997), Cherubini and Della Lunga (2001), Cherny and Madan (2009), Madan and Cherny (2010), Madan and Schoutens (2016), and Cherubini and Mulinacci (2021).†

The original contribution of our paper is to propose a two-tail probability distortion technique assigning different distortions to each tail, preserving the no-arbitrage requirements. We find that in order to avoid arbitrage, one is allowed to choose a non-additive measure (capacity), a reference probability measure (to be distorted) and one distortion function for one of the two tails: the distortion function of the other tail will be automatically recovered.

Our model is also consistent with put-call parity, which, on top of the existence of a non-additive measure, is the key requirement for the absence of arbitrage in models generating bid and ask prices (Cerreia-Vioglio *et al.* 2015, Bastianello *et al.* 2024).‡

Put-call parity implies the existence of a link between the bid-ask spread of the underlying and that of the corresponding options in order to rule out arbitrage. This emerges clearly from the basic requirements of the seminal Merton (1973) paper on the theory of rational option pricing. Denote $C^a(K)$ the ask price of a European call option to buy the underlying S at some future fixed date (which is immaterial to specify here) at the strike price K. Then, if we assume zero dividends, no arbitrage requires $C^a(0) = \pi^a(S)$, where $\pi^a(S)$ denotes the ask price of the underlying. Buying a call with a strike equal to zero is equivalent to buying the underlying. By the same token, selling a call with a strike price equal to zero, (denoted $-C^b(0)$) amounts to a short position in the underlying ($-\pi^b(S)$).

For positive strike prices K, buying a call option with strike K for $C^a(K)$ and selling a put option with the same strike, for $-P^b(K)$, is equivalent to taking a long position in the underlying asset with value $\pi^a(S)$ and borrowing the present value amount of the strike price K. From the viewpoint of the counterparty, a long position $P^a(K)$ in the put option and a short

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† This approach must not be confused with models that apply measure distortions rather than probability distortions in pure jump models (Madan *et al.* 2016, 2017, 2023).

‡ Other approaches include the seminal works of Chateauneuf *et al.* (1996) and Jouini and Kallal (1995), while Cinfrignini *et al.* (2023), Petturiti and Vantaggi (2023) recover the no-arbitrage measure in the Dempster-Shafer setting.

position $-C^b(K)$ in the call option will be equivalent to a short position $-\pi^b(S)$ in the underlying asset and an investment in the risk-free asset yielding the amount K at maturity of the option.

Putting these two results together yields the general relationship

$$C^a(K) - C^b(K) + P^a(K) - P^b(K) = \pi^a(S) - \pi^b(S) \quad (1)$$

for all strike prices K. In a no-arbitrage setting, this equality means that the sum of the bid-ask spreads of call and put options must be invariant to the strike.†

Equation (1) is a consequence of no-arbitrage restrictions, but it is not sufficient to guarantee the absence of arbitrage. In fact, while in the one-price model put and call options are integrals with respect to the same probability measure, in non-linear models the integrals are taken with respect to non-additive measures, and one has to set a relationship between the capacity used for call options prices and that applied to put. Following the seminal paper by Breeden and Litzenberger (1978), one can write the put-call parity deal replicating a long position of the underlying as

$$\int_K^{\infty} \mu^a(S_T > u)\,\mathrm{d}u - \int_0^K \mu^b(S_T \le u)\,\mathrm{d}u = \pi^a(S) - K \quad (2)$$

where μ^a and μ^b are capacities or fuzzy measures, that is non-additive measures, used for representing ask and bid prices. Likewise, a short position in the underlying can be replicated from

$$-\int_K^{\infty} \mu^b(S_T > u)\,\mathrm{d}u + \int_0^K \mu^a(S_T \le u)\,\mathrm{d}u = -\pi^b(S) + K. \quad (3)$$

A sufficient condition to ensure that $C^a(K) \ge C^b(K)$ and $P^a(K) \ge P^b(K)$ for all K is

$$\mu^a(S_T > K) \ge \mu^b(S_T > K) \quad \text{and}$$
$$\mu^a(S_T \le K) \ge \mu^b(S_T \le K)$$

which corresponds to assuming ask prices greater than bid prices for a digital option paying 1 Dollar if $S_T > K$ and one paying 1 Dollar if $S_T \le K$, respectively.

Adding the two inequalities term by term yields

$$\mu^a(S_T > K) + \mu^a(S_T \le K) \ge \mu^b(S_T > K) + \mu^b(S_T \le K).$$

Equality obviously obtains only when $\mu^a(S_T > K) + \mu^a(S_T \le K) = \mu^b(S_T > K) + \mu^b(S_T \le K) = 1$, that is when the fuzzy measure is a probability measure.

In the presence of positive bid-ask spreads the capacity representing ask prices must be super-additive and the one representing bid prices must be sub-additive, but it is easy to see that this is not enough to ensure the absence of arbitrage. In fact, consider a digital option paying 1 Dollar at time T if $S_T > K$. This contract could be bought for a market price of $\mu^a(S_T > K)$. It is also clear that one could obtain the same payoff buying a risk-free asset paying 1 Dollar at time T and selling a digital option paying 1 dollar if $S_T \le K$. Under the simplifying assumption of zero interest rate the cost of this portfolio would be: $1 - \mu^b(S_T \le K)$. In order to avoid arbitrage we must then have:

$$\mu^a(S_T > K) = 1 - \mu^b(S_T \le K), \quad \forall K.$$

This relation is implicit in (2) since can be obtained by differentiating both sides with respect to K. Similarly, from (3) we get the equation

$$\mu^b(S_T > K) = 1 - \mu^a(S_T \le K), \quad \forall K.$$

A more detailed derivation of the put-call parity relationship for digital options is addressed in Appendix A.2.

So, no-arbitrage between digital call and put options leads to a duality relationship between super-additive and sub-additive measures that is well known in the literature (see for example Klir and Folger 1992, chapter 4, and Gilboa 1989). Given a super-additive measure $\mu^a(A)$ representing ask prices, the sub-additive measure representing bid prices must be obtained from

$$\mu^b(A) = 1 - \mu^a(A^c) \quad (4)$$

where $A^c = \Omega \setminus A$ is the complement set of A. This condition on digital options is not required in one-price models, because it is trivially granted by the fact that the prices are probability measures: $1 - \mu^a(A^c) = \mu^a(A) = \mu^b(A)$.

In probability distortion models, to which we refer, the standard no-arbitrage requirements of convergence to the final payoff, horizontal and vertical no-arbitrage (Carr and Madan 2005, Davis and Hobson 2007) are actually inherited from the probability distribution that is chosen as reference. If we denote $H(K) = \mathbb{Q}(S_T \le K)$ the reference probability model, the only no-arbitrage requirement is (Jouini and Kallal 1995)

$$\mu^a(S_T \le K) \ge H(K) \ge \mu^b(S_T \le K), \quad \forall K.$$

The flexibility of these models is limited by the fact that the distortion is applied only to one tail of the distribution, as documented in Cherubini and Mulinacci (2021). In this paper, we build the dual measures μ^a and μ^b by distorting both the tails of a probability measure with cumulative distribution function H and corresponding survival function $\bar{H} \equiv 1 - H$, applying distortions $\phi(H(\cdot))$ and $\gamma(\bar{H}(\cdot))$, respectively. The put-call parity relationships in the two-tail distortion model are:

$$\int_K^{\infty} \gamma(\bar{H}(u))\,\mathrm{d}u - \int_0^K (1 - \gamma(\bar{H}(u))\,\mathrm{d}u = \pi^a(S) - K$$

and

$$-\int_K^{\infty} (1 - \phi(H(u))\,\mathrm{d}u + \int_0^K \phi(H(u))\,\mathrm{d}u = -\pi^b(S) + K.$$

† We assume perfect credit markets, that is same interest rates for lending and borrowing.

Comparing with equations (2) and (3) we obtain

$$\mu^a(S_T > u) = \gamma(\bar{H}(u)) \quad \mu^b(S_T \le u) = 1 - \gamma(\bar{H}(u)$$
$$\mu^b(S_T > u) = 1 - \phi(H(u)) \quad \mu^a(S_T \le u) = \phi(H(u).$$

This asymmetric two-tail model includes as special cases the conic finance models, recovered when $\phi = \gamma$ (Madan and Cherny 2010), and the one-tail distortion model, recovered when either $\phi(u) = u$ or $\gamma(u) = u$ (Cherubini 1997).

The plan of the paper is as follows. In section 2, we describe the two-tail distortion model. In section 3, we recover the put-call parity relationship in the general capacity-based model and in the two-tail distorted model. Section 4 proposes parametric specifications for the distortion model. A calibration exercise on stock index options is described in section 5. Section 6 concludes.

2. The model

2.1. Preliminaries and notation

Here we introduce the main concepts and tools that will be used in the two tail distortion model. Throughout the analysis, we will refer to fuzzy measures or capacities.

Definition 2.1 *Let $(\Omega, \mathcal{F})$ be a measurable space. A set function $\mu : \mathcal{F} \to [0, 1]$ is called a fuzzy measure if it satisfies the following properties:*

(1) $\mu(\emptyset) = 0$,
(2) $\mu(A) \le \mu(B)$, *whenever $A \subseteq B$, $A, B \in \mathcal{F}$,*
(3) *for every monotone sequence* $\{A_i\}_{i=1,2,\ldots} \subset \mathcal{F}$, $\lim_{i\to\infty} \mu(A_i) = \mu(\lim_{i\to\infty} A_i)$.

If $\mu(\Omega) = 1$, μ is called 'regular'.

The *dual measure* of a regular fuzzy measure μ on $(\Omega, \mathcal{F})$ is the fuzzy measure $\hat{\mu}$ defined by

$$\hat{\mu}(A) = 1 - \mu(A^c), \quad \text{for all } A \in \mathcal{F}. \tag{5}$$

We now introduce the definition of a binary operator that generalizes the classical sum among non-negative real numbers that will be used below to model super(sub)-additivity of regular fuzzy measures.

Definition 2.2 *Let I be an interval contained in $[0, +\infty)$ with $0, 1 \in I$ and $\psi : I \to [0, +\infty)$, with $\psi(0) = 0$, be a continuous, surjective and strictly increasing function. We call ψ-sum on I with generator ψ the binary operator $\oplus_\psi$ defined by*

$$x \oplus_\psi y = \psi^{-1}(\psi(x) + \psi(y)), \quad \text{for all } x, y \in I.$$

The concept of ψ-sum on $[0, +\infty)$ was proposed by Sugeno and Murofushi (1987) and extends the notion of classical sum: in fact, if $\psi = id$ in $[0, +\infty)$, then $x \oplus_{id} y = x + y$ for all $x, y \in [0, +\infty)$. It is a specific application of a general result introduced by Aczél (1966) that characterizes associative and reducible binary operators. Moreover, the notion of ψ-sum is related to that of t-conorms and we refer the reader to Cherubini and Mulinacci (2021) for a discussion on it.

By considering the restriction of ψ to the interval $[0, 1]$, we generate fuzzy measures as follows. Without loss of generality, we set $\psi(1) = 1$, since the generator of a ψ-sum is defined up to a positive multiplicative constant. Then, for whatever regular fuzzy measure assigned to disjoint sets A and B, the fuzzy measure applied to their union is

$$\mu(A \cup B) = \mu(A) \oplus_\psi \mu(B) = \psi^{-1}(\psi(\mu(A)) + \psi(\mu(B))).$$

Fuzzy measures or capacities with this property are called 'decomposable' (Chateauneuf 1996). Moreover, since the capacity is regular,

$$1 = \mu(A \cup A^c) = \psi^{-1}(\psi(\mu(A)) + \psi(\mu(A^c))). \tag{6}$$

In distortion-based models, the capacity is defined as $\mu((-\infty, a]) \equiv \phi(H(a))$, where H is a reference cumulative distribution. Since the capacity is regular, equation (6) implicitely defines a unique distortion $\mu((a, +\infty)) = \gamma(\bar{H}(a))$, where $\bar{H} = 1 - H$. We call this class of models 'two tail distortion model (2TD)'.

Based on Definition 2.2, we can introduce the associated definition of ψ difference. For $x, z \in [0, 1]$ such that $x \ge z$, we define the ψ-difference $x \ominus_\psi z$ as

$$x \ominus_\psi z = \psi^{-1}(\psi(x) - \psi(z)).$$

Definition 2.3 *Let $(\Omega, \mathcal{F})$ be a measurable space and μ be a regular fuzzy measure. μ is $\sigma - \oplus_\psi$-additive if*

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \bigoplus_{n=1}^{+\infty}{}_{\psi}\, \mu(A_n)$$

for any sequence $\{A_n\}_{n=1}^{\infty} \subset \mathcal{F}$ with $A_i \cap A_j = \emptyset$ for $i \ne j$.

$\sigma - \oplus_\psi$-additive measures have been widely considered and studied in literature (see Zhang *et al.* 2022, as one of the most recent papers on this stream of literature). From the definition, given that $\psi(1) = 1$, it immediately follows that the class of $\sigma - \oplus_\psi$-additive regular fuzzy measures coincides with the class of distorted probabilities (see the discussion about this point in Cherubini and Mulinacci 2021).

Moreover (see Proposition 2.3 in Cherubini and Mulinacci 2021), if μ is a $\sigma - \oplus_\psi$-additive regular fuzzy measure, then the dual regular fuzzy measure $\hat{\mu}$ in (5) is a $\sigma - \oplus_{\hat{\psi}}$-additive regular fuzzy measure with generator

$$\hat{\psi}(x) = \psi(1) - \psi(1 - x) = 1 - \psi(1 - x), \quad x \in [0, 1], \tag{7}$$

that we will call *dual generator*.

Given a generic regular fuzzy measure μ on a measurable space $(\Omega, \mathcal{F})$ and a random variable X, the *Choquet integral* is defined as

$$(C)\int X\, d\mu = \int_{-\infty}^{0} (\mu(X > t) - 1)\, dt + \int_{0}^{+\infty} \mu(X > t)\, dt$$

that can be rewritten as

$$(C)\int X\, d\mu = -\int_{0}^{+\infty} \hat{\mu}\left(X^- \ge s\right) ds + \int_{0}^{+\infty} \mu\left(X^+ > t\right) dt \tag{8}$$

where $X^+ = \max(0, X)$ and $X^- = \max(0, -X)$. Note that, being all involved integrands monotone functions, all the single integrals that appear in the above expressions exist: however, the resulting Choquet integral exists only when at least one of the two integrals is finite.

2.2. Construction of a suitable regular fuzzy measure

In the Introduction, we discussed that allowing for bid-ask spreads amounts to assigning a super-additive measure to call and put options ask prices and a sub-additive measure to the corresponding bid prices. We also showed that once one of the fuzzy measures is identified, the other is recovered by duality. In our probability distortion model, we assume a given reference probability measure on the real line identified by the cumulative distribution function H, and, given a generator ψ, we construct a regular $\sigma - \oplus_\psi$-additive fuzzy measure μ, by separately distorting H and $\bar{H}$, that assigns to each interval $(-\infty, x]$ and its complement $(x, +\infty)$, measures that are greater than those assigned by the reference probability measure. We prove below that this is equivalent to the existence of a pair of distortion functions which abide by the condition

$$\gamma(u) \oplus_\psi \phi(1-u) = 1, \quad \text{for all } u \in [0,1] \tag{9}$$

for a generator ψ that satisfies

$$\psi(u) + \psi(1-u) \leq 1, \quad \text{for all } u \in [0,1].$$

Formally, we consider the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and for every fuzzy measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, we set $H_\mu(x) = \mu((-\infty, x])$ and $\bar{H}_\mu(x) = \mu((x, +\infty))$, for every $x \in \mathbb{R}$. We select $H(x)$ to be a given strictly increasing cumulative distribution function on $\mathbb{R}$ and $\bar{H} = 1 - H$ the associated survival distribution function. We are going to show how to select $\phi(H(x)) \geq H(x)$ and $\gamma(\bar{H}(x)) \geq \bar{H}(x)$.

We first must show that selecting γ and ϕ distortion functions abiding by equation (9) amounts to select a unique regular $\sigma - \oplus_\psi$-additive fuzzy measure. We then start with an intermediate result based on a ψ-sum operator and the distortions discussed above.

PROPOSITION 2.1 *Let H be a strictly increasing cumulative distribution function on $\mathbb{R}$. The following are equivalent:*

(i) *There exist a generator ψ and two strictly increasing bijections $\phi : [0,1] \to [0,1]$ and $\gamma : [0,1] \to [0,1]$ that satisfy*

$$\gamma(u) = 1 \ominus_\psi \phi(1-u) \tag{10}$$

(ii) *There exists a unique regular $\sigma - \oplus_\psi$-additive fuzzy measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that*

$$H_\mu(x) = \phi(H(x)) \quad \text{and} \quad \bar{H}_\mu(x) = \gamma(\bar{H}(x)).$$

Proof (*i.* $\Rightarrow$*ii.*). By (10), we have that $\phi(1-u) \oplus_\psi \gamma(u) = 1$ for all $u \in [0,1]$, from which it follows that $\phi(H(x)) \oplus_\psi \gamma(\bar{H}(x)) = 1$ for all $x \in \mathbb{R}$. Let $a, b \in \mathbb{R}$ with $a < b$. Since

$$\begin{aligned}
\gamma(\bar{H}(a)) \ominus_\psi \gamma(\bar{H}(b)) &= \left(1 \ominus_\psi \phi(H(a))\right) \\
&\quad \ominus_\psi \left(1 \ominus_\psi \phi(H(b))\right) \\
&= \left(\psi^{-1}(1 - \psi(\phi(H(a))))\right) \\
&\quad \ominus_\psi \left(\psi^{-1}(1 - \psi(\phi(H(b))))\right) \\
&= \psi^{-1}(\psi(\phi(H(b))) \\
&\quad - \psi(\phi(H(a)))) = \\
&= \phi(H(b)) \ominus_\psi \phi(H(a)),
\end{aligned}$$

we define

$$\mu((a,b]) = \gamma(\bar{H}(a)) \ominus_\psi \gamma(\bar{H}(b)) = \phi(H(b)) \ominus_\psi \phi(H(a)).$$

μ, as a distorted probability, can be uniquely extended to a $\sigma - \oplus_\psi$-additive fuzzy measure on $\mathcal{B}(\mathbb{R})$ and satisfies, by construction, the requirements.

(*ii.* $\Rightarrow$*i.*) immediately follows from the fact that, being μ regular,

$$H_\mu(x) \oplus_\psi \bar{H}_\mu(x) = \phi(H(x)) \oplus_\psi \gamma\left(\bar{H}(x)\right) = 1, \quad \text{for all } x \in \mathbb{R}.$$

■

We now give conditions on ϕ which ensure both $\phi(u) \geq u$ and $\gamma(u) \geq u$. The following Lemma shows that this is feasible if the fuzzy measure is non-additive.

LEMMA 2.1 *Let ψ be a generator so that*

$$\psi(u) + \psi(1-u) \leq 1 \tag{11}$$

and $\phi : [0,1] \to [0,1]$ be a strictly increasing bijection. Let $\gamma : [0,1] \to [0,1]$ given by (10). *$\phi(u) \geq u$ and $\gamma(u) \geq u$ for all $u \in [0,1]$ if and only if*

$$u \leq \phi(u) \leq 1 \ominus_\psi (1-u). \tag{12}$$

Proof $\gamma(u) \geq u$ is equivalent to $\phi(u) \leq 1 \ominus_\psi (1-u)$. In fact

$$\begin{aligned}
\gamma(u) \geq u, \quad \forall u \in [0,1] &\iff \psi^{-1}(\psi(1) - \psi(\phi(1-u))) \\
&\qquad \geq u, \ \forall u \in [0,1] \\
&\iff \psi^{-1}(\psi(1) - \psi(\phi(u))) \\
&\qquad \geq 1-u, \ \forall u \in [0,1] \\
&\iff \psi(1) - \psi(1-u) \geq \psi(\phi(u)), \\
&\qquad \forall u \in [0,1] \\
&\iff 1 \ominus_\psi (1-u) \geq \phi(u), \\
&\qquad \forall u \in [0,1].
\end{aligned}$$

Notice that since $\phi(u) \geq u$ and $\gamma(u) \geq u$ is equivalent to $u \leq \phi(u) \leq 1 \ominus_\psi (1-u)$, existence of such functions ϕ and γ requires equation (11) to hold. In fact

$$\begin{aligned}
(11) &\iff \psi(u) \leq 1 - \psi(1-u) \\
&\iff u \leq 1 \ominus_\psi (1-u).
\end{aligned}$$

■

Notice that (11) is always satisfied by super-additive and, more in particular, convex generators.

REMARK 2.1 In the extreme case $\phi(u) = u$ we have $\gamma(u) = 1 \ominus_\psi (1-u)$, while in the opposite extreme case, when $\phi(u) = 1 \ominus_\psi (1-u)$, we have $\gamma(u) = u$. We refer to these cases as one-tail distortion models.

We are now ready to state the main result which is an immediate consequence of proposition 2.1 and lemma 2.1.

THEOREM 2.4 *Let H be a strictly increasing cumulative distribution function on $\mathbb{R}$. Let ψ be a generator satisfying* (11) *and $\phi : [0,1] \to [0,1]$ be a strictly increasing bijection such that*

$$u \le \phi(u) \le 1 \ominus_\psi (1-u). \tag{13}$$

Let $\gamma : [0,1] \to [0,1]$ given by (10)*. Then there exists a unique regular $\sigma - \oplus_\psi$-additive fuzzy measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that, for all $x \in \mathbb{R}$,*

$$H_\mu(x) = \phi(H(x)) \ge H(x)$$

and

$$\bar{H}_\mu(x) = \gamma(\bar{H}(x)) \ge \bar{H}(x).$$

REMARK 2.2 Symmetric results with respect to theorem 2.4 hold true if we substitute (11) with the reversed inequality:

$$\psi(u) + \psi(1-u) \ge 1 \tag{14}$$

and we will obtain $\gamma(u) \le u$ and $\phi(u) \le u$. In this case, the fuzzy measure would be sub-additive and would deliver bid prices, while in our derivation, starting from the super-additive dual we recover ask prices. In both cases, the same pair of bid and ask prices will be recovered.

REMARK 2.3 If ψ is a generator satisfying

$$\psi(u) + \psi(1-u) = 1 \tag{15}$$

then (13) implies that necessarily $\phi(u) = \gamma(u) = u$ for all $u \in [0,1]$ and $H_\mu(x) = H(x)$ and $\bar{H}_\mu(x) = \bar{H}(x)$ for all $x \in \mathbb{R}$.

2.3. The dual measure

The duality of fuzzy measures is crucial to ensure no-arbitrage of bid and ask prices. Here we complete the mathematical treatment of this duality relationship. Notice that

$$H_{\hat{\mu}}(x) = \hat{\mu}((-\infty, x]) = 1 - \mu((x, +\infty)) = 1 - \bar{H}_\mu(x)$$

and

$$\bar{H}_{\hat{\mu}}(x) = \hat{\mu}((x, +\infty)) = 1 - \mu((-\infty, x]) = 1 - H_\mu(x).$$

If μ is a regular $\sigma - \oplus_\psi$-additive fuzzy measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then we know that $\hat{\mu}$ is a regular $\sigma - \oplus_{\hat{\psi}}$-additive fuzzy measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ where $\hat{\psi}$ is the dual generator of ψ (see (7) and, more in general, section 2.1). Moreover, if $\phi : [0,1] \to [0,1]$ is a strictly increasing bijection, $\gamma : [0,1] \to [0,1]$ is given by (10) and μ is the unique regular $\sigma - \oplus_\psi$-additive fuzzy measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ given by proposition 2.1 we have that

$$H_{\hat{\mu}}(x) = \hat{\phi}(H(x)) \quad \text{and} \quad \bar{H}_{\hat{\mu}}(x) = \hat{\gamma}\left(\bar{H}(x)\right)$$

where $\hat{\phi}(u) = 1 - \gamma(1-u)$ and $\hat{\gamma}(u) = 1 - \phi(1-u)$.

Notice that if ψ satisfies (11), then $\hat{\psi}$ satisfies (14) and that, if ϕ and γ satisfy (10), then $\hat{\phi}$ and $\hat{\gamma}$ satisfy the same relation with $\hat{\psi}$ in place of ψ. Moreover, if (13) holds true, then a similar relation holds true for $\hat{\phi}$ and $\hat{\psi}$ but with reversed inequalities. Next Proposition is an immediate consequence of theorem 2.4 and remark 2.2.

PROPOSITION 2.2 *Let ψ be a generator satisfying* (11) *and $\phi : [0,1] \to [0,1]$ be a strictly increasing bijection such that*

$$u \le \phi(u) \le 1 \ominus_\psi (1-u).$$

Let $\gamma : [0,1] \to [0,1]$ given by (10)*. Then there exists a unique regular $\sigma - \oplus_\psi$-additive fuzzy measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that, for all $x \in \mathbb{R}$,*

$$H_{\hat{\mu}}(x) \le H(x) \le H_\mu(x)$$

and

$$\bar{H}_{\hat{\mu}}(x) \le \bar{H}(x) \le \bar{H}_\mu(x).$$

COROLLARY 2.5 *If $\hat{\psi} \equiv \psi$, that is if and only if the graph of ψ is symmetric around the point $(\frac{1}{2}, \frac{1}{2})$, then* (15) *holds true and, according to remark* 2.3*, $H_\mu \equiv H_{\hat{\mu}} \equiv H$ and $\bar{H}_\mu \equiv \bar{H}_{\hat{\mu}} \equiv \bar{H}$.*

3. Put-call parity for ask and bid prices

Let $(\Omega, \mathcal{F})$ be a measurable space where a regular fuzzy measure ν is defined. For the sake of simplicity, we will assume a null instantaneous interest rate ($r = 0$). Given a pay-off payable at the future date T and represented by the non-negative random variable X, its price at time 0 is

$$P(X) = (C) \int X \, \mathrm{d}\nu.$$

The representation of prices in terms of Choquet integrals for the ask price of a call option with strike price $K = 0$, which corresponds to the ask price of the underlying, can be written as

$$\pi^a(S) = \int_0^\infty \mu^a(S_T > u) \, \mathrm{d}u \equiv (C) \int_0^\infty S_T \, \mathrm{d}\mu^a$$

where μ^a is the fuzzy measure of ask prices.

Here we analyze the option pricing formulas in our 2TD model, focusing on the effect of the two distortions $\phi(\cdot)$ and $\gamma(\cdot)$. Let H be the reference cumulative distribution function of S_T with respect to an equivalent martingale probability $\mathbb{Q}$.

We assume that H is continuous and strictly increasing. The arbitrage-free reference prices of a call and a put with maturity T and strike K are given, respectively, by

$$C^{\mathbb{Q}}(K) = \int_K^{+\infty} \bar{H}(s)\,\mathrm{d}s \quad \text{and} \quad P^{\mathbb{Q}}(K) = \int_0^K H(s)\,\mathrm{d}s. \tag{16}$$

The prices of call and put options in our two-tail distortion model are obtained from the application of the Breeden and Litzenberger (1978) result using Choquet integrals of the distorted reference probability. The details of the derivation are reported in Appendix A.1. Here we report the final result.

In our probability distortion model, we denote:

$$\mu^a((0,x]) = \phi(H(x)), \quad \mu^a((x,\infty)) = \gamma(\bar{H}(x)),$$
$$\mu^b((0,x]) = 1-\gamma\left(\bar{H}(x)\right), \quad \mu^b((x,\infty)) = 1-\phi\left(H(x)\right).$$

The ask price of a call option C^a with strike K is given by

$$C^a(K) = (C)\int (S_T-K)^+ \mathrm{d}\nu = \int_K^{+\infty} \gamma(\bar{H}(s))\,\mathrm{d}s \tag{17}$$

while the bid price C^b is given by

$$\begin{aligned} C^b(K) &= -(C)\int \left[-(S_T-K)^+\right]\mathrm{d}\nu \\ &= \int_K^{+\infty} 1-\phi(H(s))\,\mathrm{d}s. \end{aligned} \tag{18}$$

As for the ask and bid prices P^a and P^b of a put with strike K we have

$$P^a(K) = (C)\int (K-S_T)^+\,\mathrm{d}\nu = \int_0^K \phi(H(s))\,\mathrm{d}s \tag{19}$$

and

$$P^b(K) = -(C)\int \left[-(K-S_T)^+\right]\mathrm{d}\nu = \int_0^K 1-\gamma(\bar{H}(s))\,\mathrm{d}s. \tag{20}$$

As a consequence of proposition 2.2 and of (17), (18), (19) and (20) we have that

$$C^a(K) \geq C^{\mathbb{Q}}(K) \geq C^b(K)$$

and

$$P^a(K) \geq P^{\mathbb{Q}}(K) \geq P^b(K),$$

that is, the prices obtained generate a bid-ask spread.

Remark 3.1 It can be easily verified that there are no spreads if and only if the dual generator $\hat{\psi}$ in (7) and ψ coincide and this fact is in line with the findings in Bastianello *et al.* (2024).

Starting from any generator ψ, the function

$$\psi_m(u) = \frac{1}{2}\psi(u) + \frac{1}{2}\hat{\psi}(u) = \frac{1}{2}\psi(u) - \frac{1}{2}\psi(1-u) + \frac{1}{2}$$

is a generator so that

$$\psi_m(u) = \hat{\psi}_m(u).$$

Since ψ is used for call prices and $\hat{\psi}$ for bid prices, ψ_m represents an average of the two measures.

However, in our model, because of corollary 2.5, bid and ask prices collapse to the arbitrage-free prices in (16).

It is now easy to verify the put-call parity relationship and the link between call and put options bid-ask prices. In fact, the ask price of S_T is

$$\begin{aligned} \pi^a(S_T) &= (C)\int S_T\,\mathrm{d}\nu \\ &= \int_0^{+\infty} \nu(S_T > u)\,\mathrm{d}u \\ &= \int_K^{+\infty} \gamma(\bar{H}(u))\,\mathrm{d}u + \int_0^K \gamma(\bar{H}(u))\,\mathrm{d}u \\ &= C^a(K) + K - \left(K - \int_0^K \gamma(\bar{H}(u))\,\mathrm{d}u\right) \\ &= C^a(K) + K - \int_0^K 1-\gamma(\bar{H}(u))\,\mathrm{d}u \\ &= C^a(K) + K - P^b(K), \end{aligned}$$

from which

$$\pi^a(S_T) - K = C^a(K) - P^b(K). \tag{21}$$

Similarly, we have that the bid price of S_T is $\pi^b(S_T) = C^b(K) + K - P^a(K)$ from which

$$\pi^b(S_T) - K = C^b(K) - P^a(K). \tag{22}$$

The option pricing formulas are reported in table 1 for the ease of the reader. They clearly recall the formulas used in conic finance. The innovation of the 2TD model is that we use different distortion functions in the two tails. No-abitrage is still guaranteed by the existence of a pricing fuzzy measure and put-call parity. The restriction implied by no-arbitrage is that the distortion functions must be the same for positions insisting on the same tail (e.g. long call and short put). Conic finance imposes a further constraint requiring that the distortion functions must be the same on the two tails and can be recovered as a particular case of the 2TD model.

Remark 3.2 Bastianello *et al.* (2024) discuss the difference between different concepts of put-call parity used in the literature. They call PCP the put-call parity version in the paper

Table 1. Option pricing formulas in the 2TD model.

Type	Call	Put
Bid	$C^b = \int_K^{+\infty} (1-\phi(H(s))\,\mathrm{d}s$	$P^b = \int_0^K (1-\gamma(1-H(s)))\,\mathrm{d}s$
Ask	$C^a = \int_K^{+\infty} \gamma(1-H(s))\,\mathrm{d}s$	$P^a = \int_0^K \phi(H(s))\,\mathrm{d}s$

by Cerreia-Vioglio *et al.* (2015). This consists of buying a call and selling a put at the money as a synthetic purchase of the underlying. In our setting, PCP reads

$$C^a(K) - P^b(K) = \pi^a(S_T) - K$$

They then compare this definition with an alternative definition emerging from Chateauneuf *et al.* (1996) that they define call-put-parity (CPP), that is defined as synthetically selling an at-the-money put option by buying a call option and selling the underlying. In our notation, CPP is

$$C^a(K) - \pi^b(S_T) + K = P^b(K).$$

It is straightforward to confirm that PCP=CPP if and only if $\pi^a(S) = \pi^b(S)$ as it is found in the paper.

4. Model specification: Sugeno model

The class of λ-fuzzy measures μ introduced by Sugeno (see, among the others, Wang and Klir 1992) is characterized by

$$\mu(A \cup B) = \mu(A) + \mu(B) + \theta\mu(A)\mu(B)$$

for all disjoint measurable sets A and B and with $\theta \in (-1, +\infty)$. As shown in Cherubini and Mulinacci (2021) this relation can be rewritten in terms of a ψ-sum as

$$\mu(A \cup B) = \mu(A) \oplus_\psi \mu(B)$$

where

$$\psi(t) = k \cdot \ln(1 + \theta t)$$

for $k > 0$. Since we have assumed the standardizing condition $\psi(1) = 1$ (see section 2.1), in the sequel we will set $k = \frac{1}{\ln(1+\theta)}$. More precisely, we consider

$$\psi(x) = \frac{\ln(1+\theta x)}{\ln(1+\theta)} \quad \theta \in (-1, 0)$$

which is a super-additive strictly increasing bijection on [0, 1]. The corresponding dual generator is

$$\hat{\psi}(x) = \frac{\ln(1+\hat{\theta}x)}{\ln(1+\hat{\theta})}, \quad \text{with } \hat{\theta} = -\frac{\theta}{\theta+1} \in (0, +\infty).$$

According to theorem 2.4, we look for distortions such that

$$u \leq \phi(u) \leq \phi_{\max}(u)$$

where $\phi_{\max}$ is computed from lemma 2.1 yielding:

$$\phi_{\max}(u) = \frac{u}{1 + \theta(1-u)}.$$

More specifically, we consider the class of distortions of the left tail of type

$$\phi_\alpha(u) = \frac{u}{1 + \alpha\theta(1-u)}, \quad \alpha \in [0, 1],$$

where α is chosen arbitrarily. In can be easily verified that $u \leq \phi_\alpha(u) \leq \phi_{\max}(u)$ for all $u \in [0, 1]$ and $\alpha \in [0, 1]$ with $\phi_0(u) = u$ and $\phi_1(u) = \phi_{\max}(u)$.

The corresponding distortion of the right tail is

$$\gamma_\alpha(u) = \frac{u(1+\theta\alpha)}{1 + \theta - \theta u(1-\alpha)},$$

with $\gamma_0(u) = \phi_{\max}(u)$ and $\gamma_1(u) = u$.

When $\alpha = 0$ we have

$$C^a(K) = \int_K^{+\infty} \frac{\bar{H}(s)}{1 + \theta\left(1 - \bar{H}(s)\right)}\, ds \quad \text{and}$$
$$C^b(K) = \int_K^{+\infty} \bar{H}(s)\, ds$$

and

$$P^a(K) = \int_0^K H(s)\, ds \quad \text{and}$$
$$P^b(K) = \int_0^K \frac{H(s)}{1 + \hat{\theta}(1 - H(s))} ds.$$

When $\alpha = 1$ we have

$$C^a(K) = \int_K^{+\infty} \bar{H}(s)\, ds \quad \text{and}$$
$$C^b(K) = \int_K^{+\infty} \frac{\bar{H}(s)}{1 + \hat{\theta}\left(1 - \bar{H}(s)\right)}\, ds$$

and

$$P^a(K) = \int_0^K \frac{H(s)}{1 + \theta(1 - H(s))}\, ds \quad \text{and}$$
$$P^b(K) = \int_0^K H(s)\, ds.$$

It can be easily verified that the distortion model allows for identical distortion of the two tails (that is $\phi_\alpha(u) = \gamma_\alpha(u)$, for all $u \in [0, 1]$) if and only if $\alpha \in (\frac{1}{2}, 1]$ and $\theta = \frac{1-2\alpha}{\alpha^2}$.

5. Calibration on market data

In this section, we show how to calibrate the model. Our goal is to fit a set of bid and ask prices of call and put options with same maturity under the restriction of the no-arbitrage requirement in the bid-ask spreads. In the discussion of the data we will see that this requirement is not always respected, but for sure it must be fulfilled in any model if the no arbitrage put-call parity relationship is satisfied. So, given a set of n strike prices we must fit $4n$ prices, and $2n$ bid-ask spreads. The theory requires that the componentwise sum of the bid-ask spreads in (21) and (22) must be a constant.

Assuming we have a standard one-price option pricing model represented by a set of parameters, $\mathbb{H}$, our task is to calibrate a set of parameters $\{\mathbb{H}, \Theta\}$ where Θ contains the parameters of the distortion functions. Denoting $\hat{C}^i(K)$ and

$\hat{P}^i(K), i = a, b$ the theoretical ask and bid prices of call and put options, and $C^i(K)$ and $P^i(K)$ the corresponding observed prices, we estimate

$$\min_{\{\mathbb{H},\Theta\}} \sum_{i=a,b} \sum_{j=1}^{n} \Big[(\hat{C}^i(K_j; \mathbb{H}, \Theta) - C^i(K_j))^2 + (\hat{P}^i(K_j; \mathbb{H}, \Theta) - P^i(K_j))^2 \Big].$$

In order to keep the model simple, throughout our analysis we will set $\Theta \equiv \{\alpha, \theta\}$ the parameters in the two-tail Sugeno distortion function of section 4.

5.1. *The reference models*

As for the choice of the cumulative distribution function H to be distorted, at first sight, one may be tempted by the standard market practice of using a lognormal model with different volatility parameters for each strike. In our setting, this amounts to define $\mathbb{H} \equiv \{\sigma_1, \ldots, \sigma_n\}$ where $\sigma_j, j = 1, \ldots, n$ are the volatility parameters in the log-normal model corresponding to each different strike. The distribution is then specified as:

$$H(y; \sigma_j) = \Phi\left(-\frac{\ln(F_t(T)/y) - 0.5\sigma_j^2(T-t)}{\sigma_j\sqrt{T-t}} \right). \tag{23}$$

Notice that instead of the spot mid-price S_t the conditional probability is a function of the forward price $F_t(T)$ for delivery at time T:

$$F_t(T) = S_t e^{(r-d)(T-t)},$$

where we allow for the risk-free rate r and the dividend yield d to be different from zero. Below we will discuss how to estimate the forward price.

Unfortunately, this parametric form, that is so rich that it can fit the market prices quite closely, is not well suited for our purpose, simply because it is not a model and equation (23) is not a distribution. It is also easy to check that the no-arbitrage requirement of invariance of the sum of the bid-ask spreads of call and put options cannot be maintained unless: $\sigma_i = \sigma, \forall i$. For this reason, we keep this as an exploratory model of the prices in order to have an idea of the distortion that could fit the bid and ask data, particularly with reference to the relative weight of the distortion in the two tails.

Once the relative weight of the distortion is estimated (which in our setting is represented by a parameter α), a proper option pricing model will be estimated. This procedure is motivated by the fact that the direct estimation of all the parameters of the model, including the α parameter, always leads this parameter to a corner solution. This may be explained by the fact that the model strives to fit the skew (that is the downward-sloping smile) concentrating the distortion on one tail only. We will see that fixing the parameter α that balances the distortion will produce model estimates that only marginally worsen the goodness of fit while fulfilling the no-arbitrage relationship.

In our empirical analysis the option pricing models applied to fit both the bid and ask prices will be the following:

- The Black and Scholes (1973) model (**BS**): the parameter of the model is the volatility of the log-normal distribution, that is assumed to be constant across the strikes. It is well known that this provides a poor fit of market option prices, particularly for short maturities. Nevertheless, here we are interested to check whether this standard model can still be useful at least for the estimation of the bid-ask spreads.
- The Carr and Torricelli (2021) model (**CT**): this model is well suited for our analysis because it builds on the put-call parity relationship. More to the point, it is based on the idea of pricing a product giving the payoff: $m_T \equiv \max(S_T, K)$ and then recovering the prices of options from put-call parity. The price of this product at time t is obtained as

$$\begin{aligned} m_t &= B_t(T) E_Q(\max(S_T, K)) \\ &= B_t(T) \left(F_t(T)^{1/b_\tau} + K^{1/b_\tau} \right)^{b_\tau}, \end{aligned} \tag{24}$$

where $B_t(T)$ is the risk-free discount factor for maturity T, $\tau \equiv T - t$ is the time to maturity and b_τ is the parameter of the model, assumed to be in the unit interval, and such that $\lim_{\tau \to 0} b_\tau = 0$.

The corresponding probability distribution implied by this model is

$$H(y) = \left(1 + \left(\frac{y}{F_t(T)} \right)^{-1/b_\tau} \right)^{b_\tau - 1}, \tag{25}$$

which is known as a specification of the Dagum distribution.
- Carr and Cherubini (2023) (**CC**) extend the CT approach to a general representation of the class of option pricing formulas endowed with the associative property. The representation exploits the property that under some technical conditions associative functions can be fully described in terms of specific functions called '*generators*'. Here we use one of the examples reported in the paper. This is obtained by using the generator: $\phi(x) = (\log(x+1))^{1/b_t}$. The value of the product paying $m_T = \max(S_T, K)$ is

$$m_t = \exp\left((\log(1+S))^{1/b_\tau} + (\log(1+K))^{1/b_\tau} \right)^{b_\tau} - 1. \tag{26}$$

The corresponding implied distribution function is:

$$H(y) = \left(\frac{\log(y+1)}{\log(m_t+1)} \right)^{1/b_\tau - 1} \frac{m_t + 1}{y+1}. \tag{27}$$

5.2. *The data*

In order to show how to calibrate the model, we investigated data for three stock index options markets. Index options markets are the most liquid, even because they are actively used to hedge single-name options. Unfortunately, since the model needs synchronous bid and ask prices applicable on the market at a given point in time, we did not have a database complying with this requirement. We then opted for downloading data in real time, once for each day, for a set of index option contracts. The data were collected every day from July 26th 2023 to September 29th for the December 2023 exercise contracts from the website of the Euronext market (live.euronext.com). The only indexes available for download on this market were the CAC40 (French market), the AEX (Dutch market) and the OBX (the Norwegian market). The option exercise is European, which makes the data set very well suited to investigate the no-arbitrage relationship and to illustrate our model. The bid and ask option prices were collected for all strikes for which both of them were simultaneously available.

In order to calibrate the model we need information about the forward price consistent with the data. Rather than extracting data on the underlying spot market and making hypotheses on the expected dividend yield we choose to estimate the forward price directly from the cross-section of call and put prices. More precisely, we first compute the forward ask price:

$$F_t^a(T) = e^{r(T-t)} \cdot \frac{1}{n} \sum_{j=1}^{n} \left(C_t^a(K_j) - P_t^b(K_j) + e^{-r(T-t)} K_j \right),$$

then the bid,

$$F_t^b(T) = -e^{r(T-t)} \cdot \frac{1}{n} \sum_{j=1}^{n} \left(P_t^a(K_j) - C_t^b(K_j) - e^{-r(T-t)} K_j \right).$$

The forward mid price is finally

$$F_t(T) = \frac{F_t^a(T) + F_t^b(T)}{2}.$$

All the data were used to compute the forward bid and ask prices for each index for each day. In the computation, we made the simplifying assumption of perfect markets, that is same interest rate r for borrowing and lending. The interest rate used is the Euribor, with the precise value for each day obtained by interpolation of the relevant tenors (1 week, 1 month, 3 and 6 months) for each maturity range.

Finally, for the estimation we use a subset of the data consisting of strike prices in the neighborhood of the *forward-at-the-money* value: generally, we restricted the analysis to levels of moneyness (defined as $K/F_t(T)$) between 90% to 110%. This choice was forced by the fact that under our specification using the entire spectrum of strikes would have increased the number of parameters of the exploratory model so much that convergence would take too much time. On top of that, the descriptive analysis of the bid and ask prices of options reported below suggests that the matter should be handled with care, since some bid and ask prices could be strategically set by the market maker at artificially high or low values. This implies that the information content of the prices may be more noisy for strikes which are far from the at-the-money value.

5.3. Option bid and ask data

Before discussing the results of the estimates it is interesting to provide a descriptive overview of the options bid and ask prices data. From the data, it emerges that the no-arbitrage relationship linking the bid-ask prices of call and put is not always borne out by the data. Only in the example of the French market we find that the sum of the bid-ask spreads is about the same across all the strike prices. For the other two markets, we find different patterns. The reason is that in a quote-driven market the market maker has always the choice to 'close' the market on one side or the other, setting the ask prices artificially high or the bid prices artificially low.

Figures 1–3 illustrate the differences in the typical behavior of the markets on data downloaded in two days of the sample. In each figure, we report the bid-ask spread of call and put options against each strike and their sum, that according to our sufficient condition for no-arbitrage should be constant. As we said, figure 1 testifies that the market for options on the CAC40 index by and large complies with the requirement. The sum of the bid-ask spread of put and call options looks constant around a mean of 12.0275 index points with a standard deviation of the estimate of 0.82.

The picture is different if we look at the Dutch market. In this case, we reported the graph for the third day because it is more symmetric and depicts a setting that is standard throughout the whole sample. Figure 2 shows that the sum of call and put options is very different around the forward-ATM value, with respect to OTM and ITM options. There exists a sort of valley in a moneyness region around 10% of the forward price in which the sum of the spread is flat around 0.6-0.7 index points, which raises to around 1.7 for the other strikes. The inspection of the call and put spreads explains that these higher values of the spreads are due to a sort of cliff of the ask prices for both call and put options. If we believe that most of the action in the market is around the ATM strike, this evidence suggests that the market makers keep the ask prices of ITM call and put options artificially high. For this reason, in this case, it is advisable to use the data around the ATM region, even though this comes at the cost of a reduction of the dimension of the sample. It is also worth noting that in the end of the sample, when the contracts get closer to maturity, the CAC40 data also take on the same configuration.

Finally, the OBX market provides a picture that is totally different from the other two. Figure 3 shows that in this market the typical pattern of the sum of call and put bid-ask prices with respect to moneyness is a triangle with an upper vertex corresponding to the forward-ATM value. Inspection of the call and put spread schedules shows that they are absolutely flat and equal (at a level of 3.75 index points in the specific day of the picture) for ITM and ATM call and put options (which explains the peak of the sum at 7.5 index points). The spread decreases for the OTM call and put options. In this situation, it is difficult to check where the value of the underlying spread could be. All we can say in this case is that the no-arbitrage bid-ask spread is somewhere between the sum at the OTM extremes (about 5 index points) and the value at the peak (7.5 index points). While the contract gets closer to maturity, the triangular shape tends to fade away and the data conform more closely to the no-arbitrage requirement.

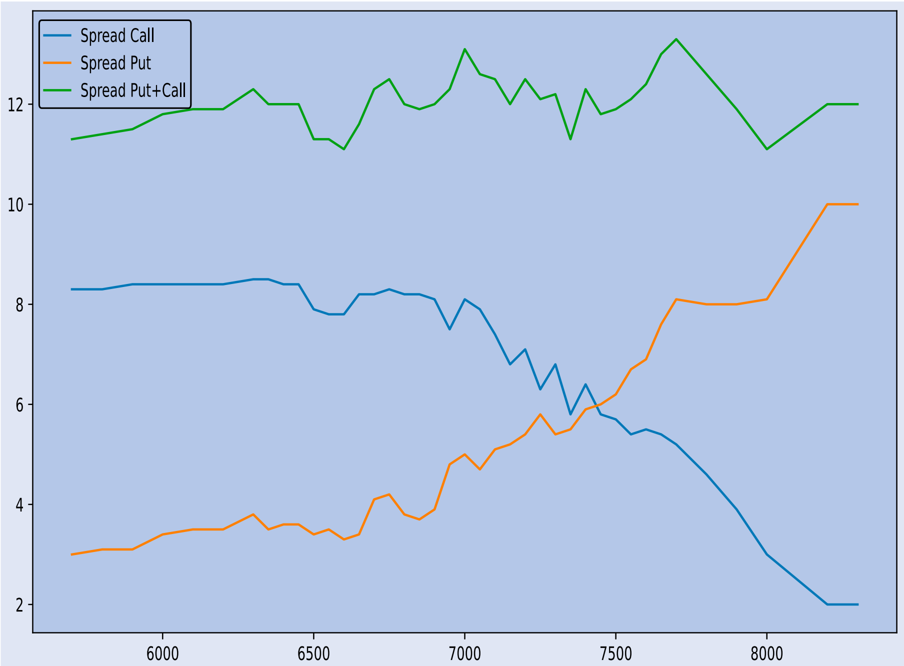

Figure 1. Bid-ask spreads of options on the CAC40 index in a snapshot of the market on July 26th 2023.

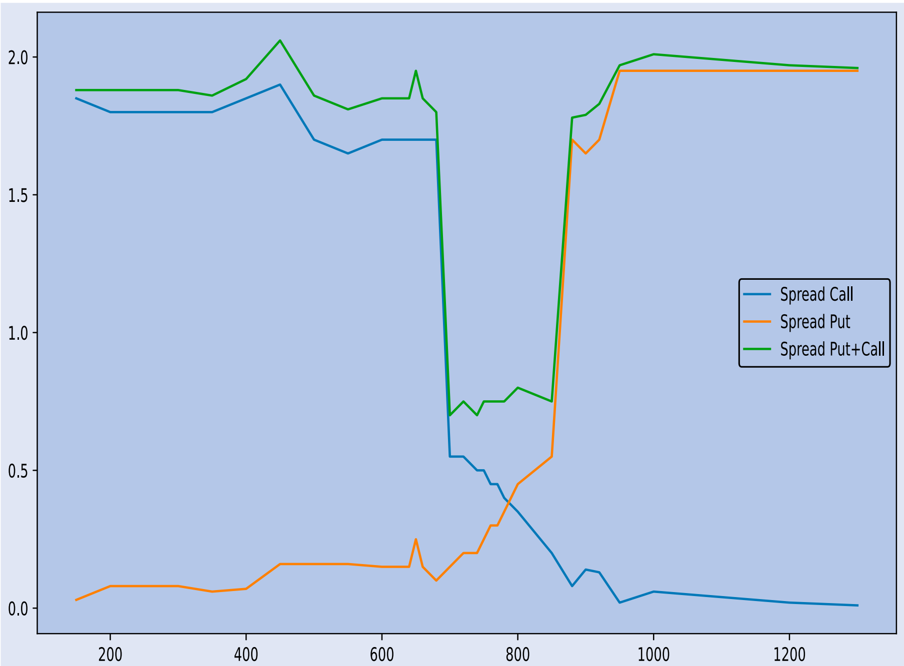

Figure 2. Bid-ask spreads of options on the AEX index in a snapshot of the market on July 28th 2023.

These examples advise caution in choosing the set of prices that ought to be used for calibration and induces a warning on possible biases that could affect the results.

5.4. Results

Here we discuss the results. First, we go over the exploratory model, with the goal to focus on the values of the estimates of the parameters α and θ. The huge flexibility of the exploratory model to allow a wide variety of smile shapes may enable to obtain a reliable estimate of the parameters, particularly parameter α. As a matter of fact, preliminary estimates of the other models, BS, CT and CC, yielded the one-tail model in a large majority of cases, producing a bad fit of the prices and an even worse fit of the bid-ask spreads. The estimate of the exploratory model provides a good fit of the prices, but still a bad fit of the bid-ask prices: the sum of the call and put bid-ask prices is decreasing with the strike.

In order to take advantage of the merits of the exploratory model and the other models we use the former to specify the α parameter and the latter to calibrate the θ parameter.

5.4.1. The exploratory model. We report here the results of our daily estimates for the exploratory model. Statistics of the

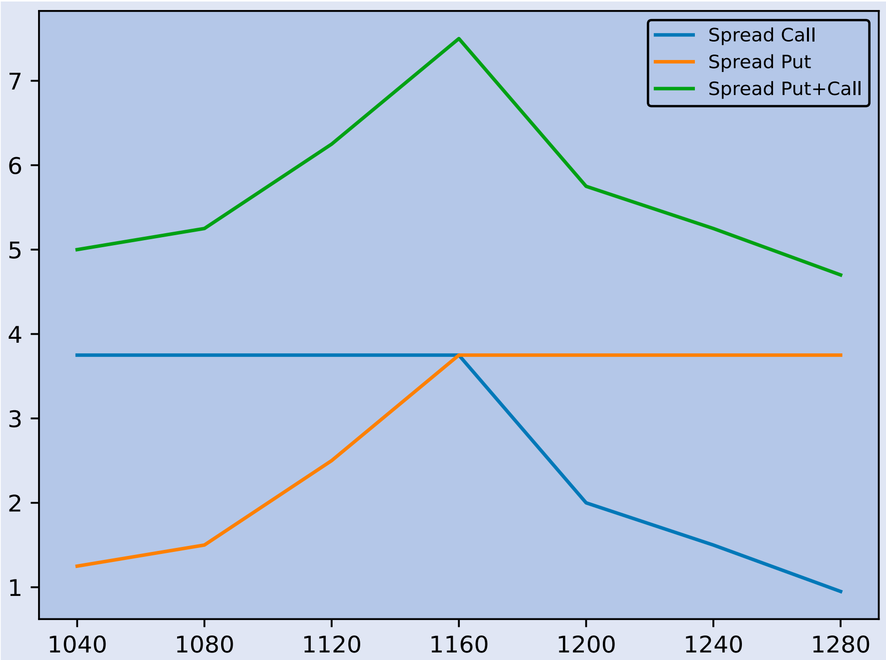

Figure 3. Bid-ask spreads of options on the OBX index in a snapshot of the market on July 26th 2023.

Table 2. Estimates of the α parameter in the exploratory model.

Statistics	CAC40	OBX	AEX
Maximum	0.559248938	0.522348722	0.666524466
3rd Quartile	0.505838434	0.520542498	0.576086468
Median	0.500020185	0.519421726	0.542586662
Mean	0.50280432	0.518977121	0.537341676
1st Quartile	0.497070238	0.517351786	0.497852298
Minimum	0.489193104	0.513926069	0.38494874

α parameter estimates are reported in table 2. For the CAC40 and the OBX market, the estimates are contained in a very narrow range and centered around 0.5. The median value is exactly 0.5 for the CAC40 and 0.52 for the OBX. Moreover, 50% of the estimates are between 0.497 and 0.506 for CAC40 and between 0.517 and 0.520. For these two markets, the 2TD model dominates the one-tail distortion model. Only in one day out of more than 90 we find an estimate of α equal to 0. The estimates are much more unstable for the AEX model. In this market, the median is 0.54 and the inter-quartile range is wider, particularly on the upper side: while the lower quartile is the same as that of the CAC40 market (0.497) the upper quartile is 0.576.

The estimates of θ are also quite stable, with median values -0.02709 (and inter-quartile range between -0.02899 and -0.02585) for the CAC40 index and -0.08423 for the OBX market (quartiles value from -0.08759 and -0.07978). Estimates are again more unstable for the AEX market, where the median estimate is closer to zero (-0.01011) and the inter-quartile range is wider (from -0.01518 to 0.00201).

5.4.2. The three models. Here we report the results obtained for the three models taken into consideration, namely the Black and Scholes one (**BS**), Carr and Torricelli (**CT**) and Carr and Cherubini (**CC**). For each market, tables 3–5 report the statistics of the parameter θ for the three models, and that from the exploratory for comparison. A common feature of the results is that the estimate of the parameter is very similar across the three models. So, the choice of the reference model is not very relevant for the specification of the tail distortion parameter. A comparison with the estimates obtained in the explanatory model shows that the three models produce values that are lower than those obtained in the exploratory model, even though generally the difference is not substantial. This would suggest that reference models that do not represent the smile (the skew, to be precise) properly provide (moderately) higher estimates of the distortion parameter.

In figures 4 and 5 we also report the dynamics of the estimates of θ for two of the indexes, CAC40 and OBX. For the AEX index it was not possible to produce a meaningful graph because the estimate of the parameter is too volatile, being mostly affected by the instability of the α parameter estimates. Figure 4 reports that the estimate of the parameter is quite stable, except for two days in the sample in which all the models, including the explanatory one, show a downward jump. As for the difference, between the explanatory model and the three models we see that the positive difference emerges after August 10th. In the first part of the sample, the difference is also negative. In figure 5, we observe that for the OBX index the estimates of the parameter are less stable and seem to follow a downward trend while the time to maturity of the contract shrinks. The difference between estimates of the parameter in the exploratory and the other models looks more hefty.

It must be noticed that we have much more data for calibration for the CAC40 index than for the OBX. For each day, we have data for about 25 strike prices of the CAC40 index, compared to about 10 for the OBX. This would lead to interpreting the results as more reliable for the former than for the latter. The same argument is confirmed if we check the consistency of the estimates of the bid-ask spreads of call and put options

Table 3. Estimates of the θ parameter: CAC40 index.

Statistics	Exploratory	BS	CT	CC
Maximum	– 0.020936124	– 0.016994662	– 0.017202864	– 0.016935279
3rd Quartile	– 0.025853851	– 0.027740595	– 0.027623804	– 0.027657764
Median	– 0.027086368	– 0.030533086	– 0.030422971	– 0.030478151
Mean	– 0.02774781	– 0.030431234	– 0.030373217	– 0.030446915
1st Quartile	– 0.028991283	– 0.032688007	– 0.032667949	– 0.03270937
Minimum	– 0.041724605	– 0.044994622	– 0.044911944	– 0.044918301

Table 4. Estimates of the θ parameter: AEX index.

Statistics	Exploratory	BS	CT	CC
Maximum	0.0	0.0	0.0	0.0
3rd Quartile	– 0.002013783	– 0.001145225	– 0.002233888	– 0.002013783
Median	– 0.010105817	– 0.009946715	– 0.009884779	– 0.010105817
Mean	– 0.009316295	– 0.009270118	– 0.009328839	– 0.009316295
1st Quartile	– 0.01517962	– 0.015437528	– 0.015406763	– 0.01517962
Minimum	– 0.033787571	– 0.036800369	– 0.03329237	– 0.033787571

with the implied forward spread derived from the no-arbitrage requirement. Figures 6 and 7 report the sum of call and put bid-ask spreads produced by the estimates of the model compared with the same average figure computed from the raw data. Figure 6 shows that the three models fit the implied forward bid-ask spread observed quite closely, except perhaps in the first part of the sample. For the OBX index figure 7 shows that the three models constantly undervalue the implied forward spread, even though the dynamics is similar. Again, the evidence for AEX is not reported because the estimates are much too unstable to produce reliable results. Overall, the no-arbitrage constraint required by put-call parity is verified only in the CAC40 case, it is only partially verified for the OBX case, while is not respected at all in the AEX case.

Table 5. Estimates of the θ parameter: OBX index.

Statistics	Exploratory	BS	CT	CC
Maximum	– 0.075807264	– 0.079605946	– 0.079801317	– 0.079888134
3rd Quartile	– 0.07977822	– 0.082488145	– 0.082758082	– 0.082897613
Median	– 0.084226495	– 0.087391932	– 0.087308595	– 0.087402376
Mean	– 0.084354536	– 0.087556961	– 0.087902913	– 0.088040824
1st Quartile	– 0.087586488	– 0.090948496	– 0.091344693	– 0.091500947
Minimum	– 0.097652302	– 0.101086632	– 0.101494039	– 0.10162891

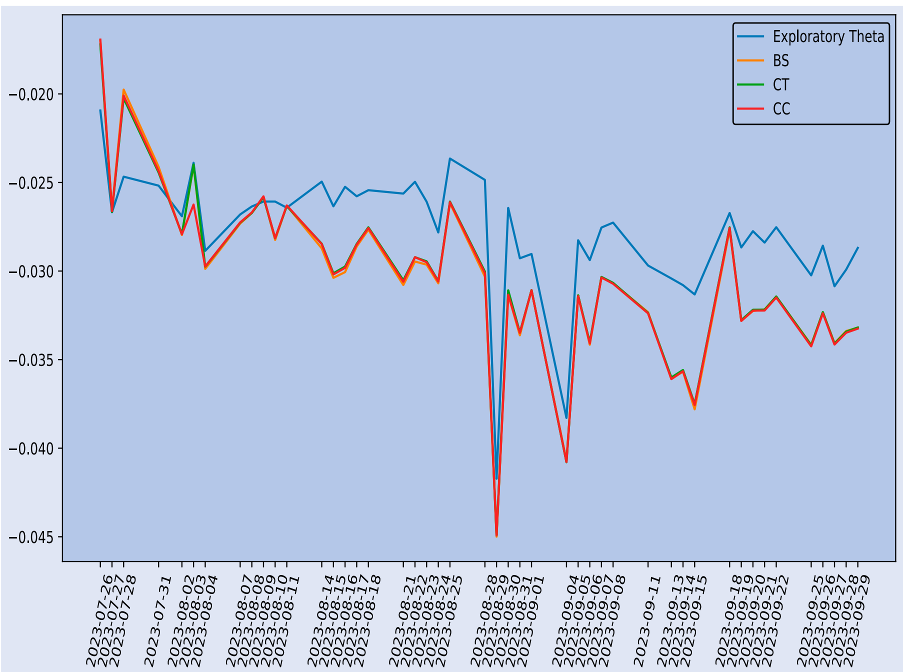

Figure 4. θ estimates for the CAC40 index: July 26th-September 29th.

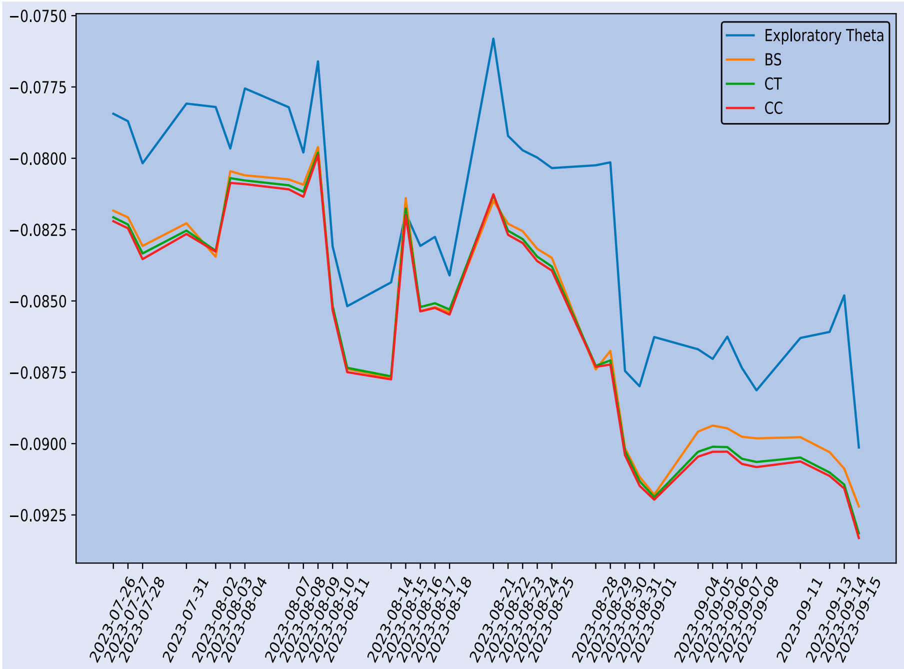

Figure 5. θ estimates for the OBX index: July 26th-September 15th.

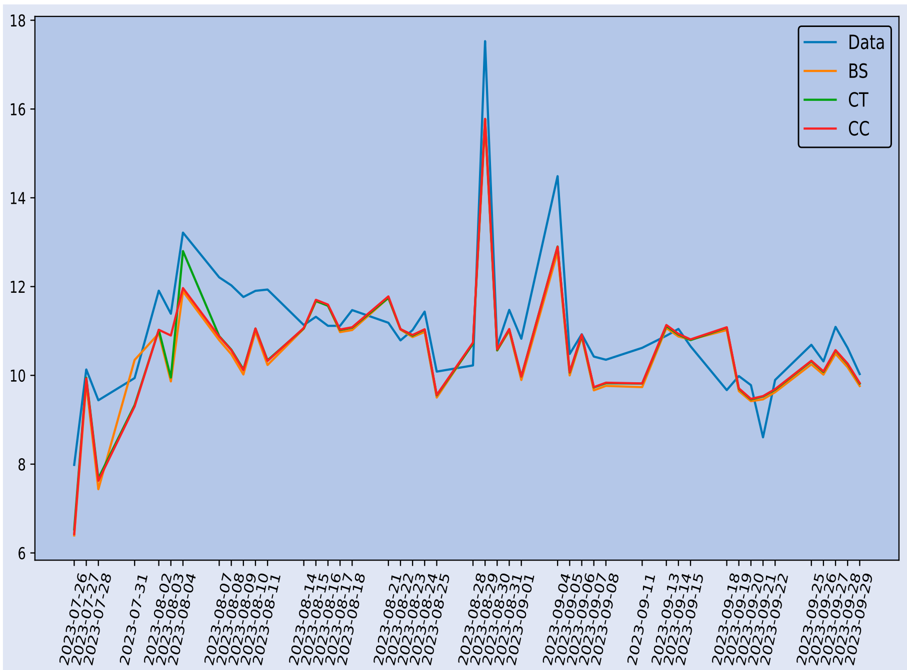

Figure 6. Comparison of the estimated sum of call and put bid-ask spreads and the average sum observed in the market: CAC40 index: July 26th-September 29th.

6. Conclusion

In this paper, we propose a way to generating and fitting bid-ask spreads of call and put options by applying a distortion to each tail. The two distortion functions are linked by a super-additive measure. More formally, we select a super-additive measure

$$\psi(u) + \psi(1-u) \leq 1$$

and two distortion functions $\gamma(x)$ and $\phi(x)$ such that:

$$\psi(\gamma(x)) + \psi(\phi((1-u)) = 1.$$

The distortion functions are applied to a reference probability distribution $H(x)$. The family of one-tail distortion (1TD) that has been typically applied until now is obtained by setting either $\gamma(x) = x$ or $\phi(x) = x$. Call and put bid and ask prices are obtained by applying the Choquet integral to the distorted measure. The prices obtained are consistent with the

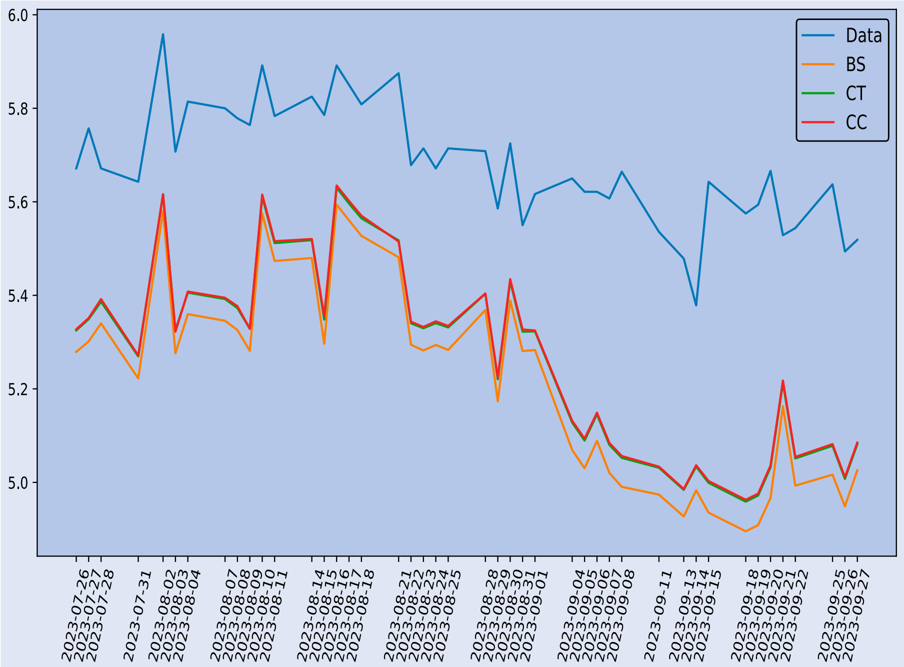

Figure 7. Comparison of the estimated sum of call and put bid-ask spreads and the average sum observed in the market: OBX index: July 26th-September 27th.

put-call parity, which is known to be the condition required for arbitrage-free pricing in pricing models based on capacities. The put-call parity condition requires that the sum of call and put bid-ask spreads be constant across the strike prices for every maturity.

We calibrate the model on three stock index options with European exercise, the CAC40 (France), AEX (the Netherlands) and OBX (Norway). We estimate a two-tail extension of the Sugeno fuzzy measure, enriched by a parameter that weighs the distortion in the two tails ($\gamma(\cdot)$ and $\phi(\cdot)$). In a two-step estimation technique, we first estimate the relevance of this parameter using an exploratory model which applies the Black and Scholes pricing formula with a different volatility parameter for each strike. We found overwhelming evidence in favor of the 2TD, with the weight parameter very close to 0.5 in two of the three option indexes (CAC40 and OBX), and the result was also confirmed for the AEX contract, even if the calibration was much less stable. Since the exploratory model violates the put-call parity condition, in the second calibration step we evaluate the distortion parameter using three full-fledged option pricing models, keeping the same relative weight of the tail distortion. The tail distortion parameter turns out to be amazingly similar across the different models, and also quite in line with the one estimated in the exploratory model. Again, only the results for the AEX index display some instability problem.

On a final note, our results point out that the 2TD solution generally dominates the 1TD one, being more flexible. In two option index contracts out of three, the model also turns out to be remarkably stable. Further research will address the estimation of the model with more flexible distortions and reference models that are more well suited to represent the implied volatility smile.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

References

Aczél, J., *Lectures on Functional Equations and Their Applications*, 1966 (Academic Press: New York).

Bastianello, L., Chateauneuf, A. and Cornet, B., Put-call parities, absence of arbitrage opportunities and non linear pricing rules. *Math. Finance*, 2024, **34**(4), 1087–1327.

Black, F. and Scholes, M., The pricing of options and corporate liabilities. *J. Polit. Econ.*, 1973, **81**(3), 637–654.

Breeden, D.T. and Litzenberger, R.H., Prices of state contingent claims implicit in option prices. *J. Bus.*, 1978, **51**(4), 621–651.

Carr, P. and Cherubini, U., Option pricing generators. *Front. Math. Finance*, 2023, **2**, 150–169.

Carr, P. and Madan, D., A note on sufficient conditions for no arbitrage. *Fin. Res. Lett.*, 2005, **2**, 125–130.

Carr, P. and Torricelli, L., Additive logistic processes in option pricing. *Finance Stoch.*, 2021, **25**, 689–724.

Cerreia-Vioglio, S., Maccheroni, F. and Marinacci, M., Put-Call parity and market frictions. *J. Econ. Theory.*, 2015, **157**, 730–762.

Chateauneuf, A., Decomposable capacities, distorted probabilities and concave capacities. *Math. Soc. Sci.*, 1996, **31**(1), 19–37.

Chateauneuf, A., Kast, R. and Lapied, A., Choquet pricing for financial mar- kets with frictions. *Math. Finance*, 1996, **6**(3), 323–330.
Cherny, A. and Madan, D.B., New measures for performance evaluation. *Rev. Financ. Stud.*, 2009, **22**, 2571–2606.
Cherubini, U., Fuzzy measures and asset prices: Accounting for information ambiguity. *Appl. Math. Finance*, 1997, **4**(3), 135–149.
Cherubini, U. and Della Lunga, G., Liquidity and credit risk. *Appl. Math. Finance*, 2001, **8**(2), 79–95.
Cherubini, U. and Mulinacci, S., Extensions and distortions of λ-fuzzy measures. *Fuzzy Sets Syst.*, 2021, **412**, 27–40.
Cinfrignini, A., Petturiti, D. and Vantaggi, B., Envelopes of equivalent martingale measures and a generalized no-arbitrage principle in a finite setting. *Ann. Oper. Res.*, 2023, **321**, 103–137.
Davis, M.H.A. and Hobson, D.G., The range of traded option prices. *Math. Finance*, 2007, **17**, 1–14.
Gilboa, I., Duality in non-additive expected utility theory. *Ann. Oper. Res.*, 1989, **19**, 405–414.
Jouini, E. and Kallal, K., Martingales and arbitrage in securities markets with transaction costs. *J. Econ. Theory.*, 1995, **66**(1), 178–197.
Klir, G.J. and Folger, T.A., *Fuzzy Sets, Uncertainty, and Information*, 1992 (Prentice Hall: Singapore).
Madan, D.B. and Cherny, A., Markets as a counterparty: An introduction to conic finance. *Int. J. Theor. Appl. Finance*, 2010, **13**(8), 1149–1177.
Madan, D.B. and Schoutens, W., *Applied Conic Finance*, 2016 (Cambridge University Press: Cambridge, UK).
Madan, D.B., Pistorius, M. and Stadje, M., Convergence of BSDΔEs driven by random walks to BSDEs: The case of (In)Finite activity jumps with general driver. *Stoch. Process. Their. Appl.*, 2016, **126**, 1553–1584.
Madan, D.B., Pistorius, M. and Stadje, M., On dynamic spectral risk measures, a limit theorem and optimal portfolio allocation. *Finance Stoch.*, 2017, **21**, 1073–1102.
Madan, D.B., Schoutens, W. and Wang, K., Option returns. *Front. Math. Finance*, 2023, **2**(2), 244–264.
Merton, R.M., The theory of rational option pricing. *Bell J. Econ. Manage. Sci.*, 1973, **4**(1), 141–183.
Petturiti, D. and Vantaggi, B., No-Arbitrage pricing with α-DS mixtures in a market with bid-ask spreads. *Proc. Mach. Learn. Res.*, 2023, **215**, 401–411. ISIPTA 2023.
Sugeno, M. and Murofushi, T., Pseudo-additive measures and integrals. *J. Math. Anal. Appl.*, 1987, **122**, 197–222.
Wang, Z. and Klir, G.J., *Fuzzy Measure Theory*, 1992 (Springer Science: New York).
Zhang, D., Mesiar, R. and Pap, E., Pseudo-integral and generalized Choquet integral. *Fuzzy Sets Syst.*, 2022, **446**, 193–221.

Appendix

A.1. Call and put prices with the 2TD model

Here we derive the bid and ask prices of call and put options in the 2TD model. We remind:

$$\mu^a(B) = \nu\,(S_T \in B)\,, \quad \text{for all } B \in \mathcal{B}(\mathbb{R})$$

for ask prices and

$$\mu^b(B) = \hat{\nu}\,(S_T \in B)\,, \quad \text{for all } B \in \mathcal{B}(\mathbb{R})$$

for bid prices.

The strategy of the proof is as follows. For call and put ask prices we consider the fuzzy measure ν while for the corresponding bid prices, we use the associated dual measure $\hat{\nu}$. We remind that measure ν is associated to the distorted probability H_μ defined in proposition 2.1 while $\hat{\nu}$ is associated to the distorted probability $H_{\hat{\mu}}$ defined in proposition 2.2.

Formally, ask and bid prices are represented in terms of H_μ and $H_{\hat{\mu}}$ as follows

$$\mu^a((0,x]) = H_\mu(x), \quad \mu^a((x,\infty)) = \bar{H}_\mu(x),$$
$$\mu^b((0,x]) = H_{\hat{\mu}}(x), \qquad \mu^b((x,\infty)) = \bar{H}_{\hat{\mu}}(x).$$

The ask price C^a of a call with strike K is given by

$$\begin{aligned} C^a(K) &= (C)\int (S_T - K)^+ \, \mathrm{d}\nu \\ &= \int_0^{+\infty} \nu\left((S_T - K)^+ > t\right)\, \mathrm{d}t \\ &= \int_0^{+\infty} \nu\,(S_T > t + K)\, \mathrm{d}t \\ &= \int_0^{+\infty} \bar{H}_\mu(t+K)\, \mathrm{d}t \\ &= \int_K^{+\infty} \bar{H}_\mu(s)\, \mathrm{d}s \end{aligned} \tag{A1}$$

while the bid price C^b is given by

$$\begin{aligned} C^b(K) &= -(C)\int \left[-(S_T - K)^+\right] \mathrm{d}\nu \\ &= \int_0^{+\infty} \hat{\nu}\left((S_T - K)^+ \geq t\right)\, \mathrm{d}t \\ &= \int_0^{+\infty} \hat{\nu}\,(S_T \geq t + K)\, \mathrm{d}t \\ &= \int_0^{+\infty} \bar{H}_{\hat{\mu}}(t+K)\, \mathrm{d}t \\ &= \int_K^{+\infty} \bar{H}_{\hat{\mu}}(s)\, \mathrm{d}s. \end{aligned} \tag{A2}$$

As for the ask and bid prices P^a and P^b of a put with strike K we have

$$\begin{aligned} P^a(K) &= (C)\int (K - S_T)^+ \, \mathrm{d}\nu \\ &= \int_0^{+\infty} \nu\left((K - S_T)^+ > t\right)\, \mathrm{d}t \\ &= \int_0^{+\infty} \nu\,(S_T < K - t, S_T < K)\, \mathrm{d}t \\ &= \int_0^{K} H_\mu(K - t)\, \mathrm{d}t \\ &= \int_0^{K} H_\mu(s)\, \mathrm{d}s \end{aligned} \tag{A3}$$

and

$$\begin{aligned} P^b(K) &= -(C)\int \left[-(K - S_T)^+\right] \mathrm{d}\nu \\ &= \int_0^{+\infty} \hat{\nu}\left((K - S_T)^+ \geq t\right)\, \mathrm{d}t \\ &= \int_0^{K} H_{\hat{\mu}}(s)\, \mathrm{d}s. \end{aligned} \tag{A4}$$

A.2. Put-call parity for digital options

Here we describe the no-arbitrage relationship between digital call and put options, and we show the relationship to the duality between measures. We denote $DC(K)$ the price of a digital option paying 1 euro if at time T the underlying S_T is higher to the strike price K and

$DP(K)$ the price of the corresponding digital put option, that is the option paying 1 euro if it will be below. We carry out the proof from the financial viewpoint of its cost of production. We know that in a one-price market the digital products are defined as

$$DC(K) = \lim_{h\to 0} \frac{C(K) - C(K+h)}{h}$$

and

$$DP(K) = \lim_{h\to 0} \frac{P(K+h) - P(K)}{h}.$$

In financial terms, this denotes the limit of trading large amounts $1/h$ of call and put spreads of strikes of distance h. It is immediate to see that the limit does not carry over if we allow for bid-ask spreads. In fact, assuming a deal of a finite amount $1/h$ ($h > 0$) of call and put spreads we have that the ask price is

$$\begin{aligned} DC^a(K) &= \frac{C^a(K) - C^b(K+h)}{h} \\ &= \frac{C^b(K) - C^b(K+h)}{h} + \frac{C^a(K) - C^b(K)}{h} \end{aligned}$$

and

$$\begin{aligned} DP^a(K) &= \frac{P^a(K+h) - P^b(K)}{h} \\ &= \frac{P^a(K+h) - P^a(K)}{h} + \frac{P^a(K) - P^b(K)}{h}. \end{aligned}$$

We now use the dual relationships proved in the paper to obtain:

$$\begin{aligned} C^b(K) &= \int_K^\infty (1 - \phi(H(s)))\, \mathrm{d}s \\ P^a(K) &= \int_0^K \phi(H(s))\, \mathrm{d}s \end{aligned}$$

and the no-arbitrage restriction

$$C^a(K) - C^b(K) + P^a(K) - P^b(K) = \pi^a(S_T) - \pi^b(S_T)$$

to obtain:

$$\begin{aligned} &DP^a(K) + DC^a(K) \\ &\quad = \frac{P^a(K+h) - P^a(K)}{h} - \frac{C^b(K+h) - C^b(K)}{h} \\ &\qquad + \frac{\pi^a(S_T) - \pi^b(S_T)}{h} \\ &\quad = \frac{\int_K^{K+h} \phi(H(s))\, \mathrm{d}s}{h} + \frac{\int_K^{K+h} (1 - \phi(H(s)))\, \mathrm{d}s}{h} \\ &\qquad + \frac{\pi^a(S_T) - \pi^b(S_T)}{h} \\ &\quad = 1 + \frac{\pi^a(S_T) - \pi^b(S_T)}{h}. \end{aligned}$$

Following the same reasoning, if we consider the mirroring portfolios, we get

$$\begin{aligned} DC^b(K) &= -\left[\frac{C^a(K+h) - C^a(K)}{h} + \frac{C^a(K) - C^b(K)}{h}\right], \\ DP^b(K) &= -\left[-\frac{P^b(K+h) - P^b(K)}{h} + \frac{P^a(K) - P^b(K)}{h}\right] \end{aligned}$$

and

$$\begin{aligned} &DP^b(K) + DC^b(K) \\ &\quad = -\left[-\frac{P^b(K+h) - P^b(K)}{h} + \frac{C^a(K+h) - C^a(K)}{h}\right. \\ &\qquad \left. + \frac{\pi^a(S_T) - \pi^b(S_T)}{h}\right] \\ &\quad = \frac{\int_K^{K+h} \left(1 - \gamma(\bar{H}(s))\right) \mathrm{d}s}{h} + \frac{\int_K^{K+h} \gamma(\bar{H}(s))\, \mathrm{d}s}{h} \\ &\qquad - \frac{\pi^a(S_T) - \pi^b(S_T)}{h} \\ &\quad = 1 - \frac{\pi^a(S_T) - \pi^b(S_T)}{h} \end{aligned}$$

and the aggregated bid-ask spread is

$$\begin{aligned} &DP^a(K) + DC^a(K) - \left(DP^b(K) + DC^a(K)\right) \\ &\quad = 2\frac{\pi^a(S_T) - \pi^b(S_T)}{h}. \end{aligned}$$

We conclude that:

- the no-arbitrage relationship $DC(K) + DP(K) = 1$ holds only if the bid-ask spread of the underlying is zero;
- the no-arbitrage relationship requires the duality between the distorted measure used for the call and that used for the put option.